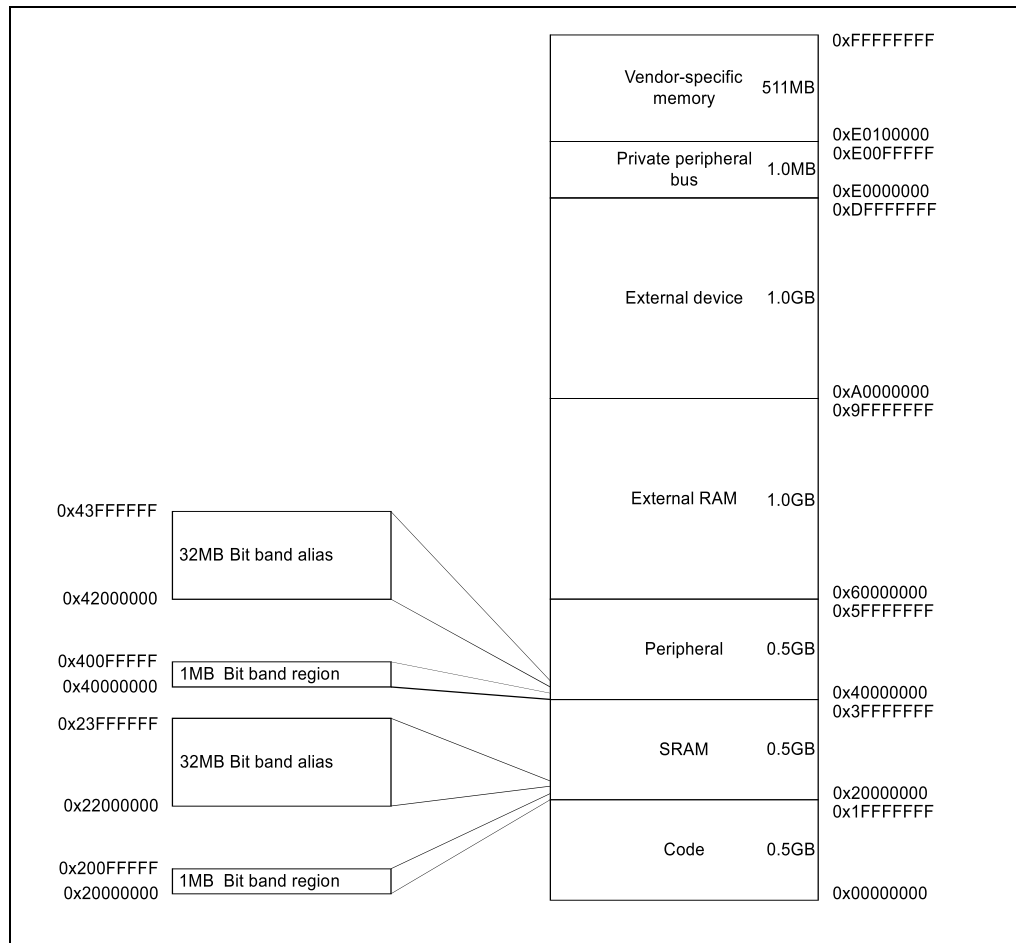


2.2 Memory model

This section describes the processor memory map, the behavior of memory accesses, and the bit-banding features. The processor has a fixed memory map that provides up to 4 GB of addressable memory.

Figure 8. Memory map



The regions for SRAM and peripherals include bit-band regions. Bit-banding provides atomic operations to bit data, see [Section 2.2.5: Bit-banding on page 32](#).

The processor reserves regions of the *Private peripheral bus* (PPB) address range for core peripheral registers, see [Section 4.1: About the STM32 Cortex-M4 core peripherals on page 193](#).

2.2.5 Bit-banding

A bit-band region maps each word in a *bit-band alias* region to a single bit in the *bit-band region*. The bit-band regions occupy the lowest 1 Mbyte of the SRAM and peripheral memory regions.

The memory map has two 32 Mbyte alias regions that map to two 1 Mbyte bit-band regions:

- Accesses to the 32 Mbyte SRAM alias region map to the 1 Mbyte SRAM bit-band region, as shown in [Table 14](#)
- Accesses to the 32 MB peripheral alias region map to the 1 MB peripheral bit-band region, as shown in [Table 15](#).

Table 14. SRAM memory bit-banding regions

Address range	Memory region	Instruction and data accesses
0x20000000-0x200FFFFF	SRAM bit-band region	Direct accesses to this memory range behave as SRAM memory accesses, but this region is also bit addressable through bit-band alias.
0x22000000-0x23FFFFFF	SRAM bit-band alias	Data accesses to this region are remapped to bit band region. A write operation is performed as read-modify-write. Instruction accesses are not remapped.

Table 15. Peripheral memory bit-banding regions

Address range	Memory region	Instruction and data accesses
0x40000000-0x400FFFFF	Peripheral bit-band region	Direct accesses to this memory range behave as peripheral memory accesses, but this region is also bit addressable through bit-band alias.
0x42000000-0x43FFFFFF	Peripheral bit-band alias	Data accesses to this region are remapped to bit-band region. A write operation is performed as read-modify-write. Instruction accesses are not permitted.

Note: A word access to the SRAM or peripheral bit-band alias regions map to a single bit in the SRAM or peripheral bit-band region.

Bit band accesses can use byte, halfword, or word transfers. The bit band transfer size matches the transfer size of the instruction making the bit band access.

The following formula shows how the alias region maps onto the bit-band region:

$$\text{bit_word_offset} = (\text{byte_offset} \times 32) + (\text{bit_number} \times 4)$$

$$\text{bit_word_addr} = \text{bit_band_base} + \text{bit_word_offset}$$

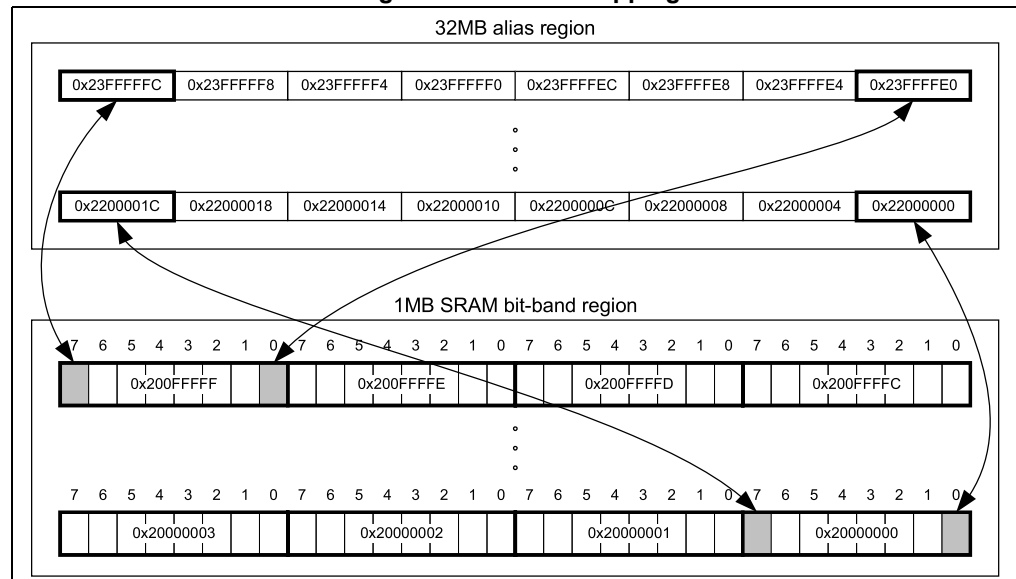
Where:

- Bit_word_offset is the position of the target bit in the bit-band memory region.
- Bit_word_addr is the address of the word in the alias memory region that maps to the targeted bit.
- Bit_band_base is the starting address of the alias region.
- Byte_offset is the number of the byte in the bit-band region that contains the targeted bit.
- Bit_number is the bit position, 0-7, of the targeted bit.

Figure 9 on page 33 shows examples of bit-band mapping between the SRAM bit-band alias region and the SRAM bit-band region:

- The alias word at 0x23FFFFED maps to bit[0] of the bit-band byte at 0x200FFFFF: $0x23FFFFED = 0x22000000 + (0xFFFF*32) + (0*4)$.
- The alias word at 0x23FFFFFC maps to bit[7] of the bit-band byte at 0x200FFFFF: $0x23FFFFFC = 0x22000000 + (0xFFFF*32) + (7*4)$.
- The alias word at 0x22000000 maps to bit[0] of the bit-band byte at 0x20000000: $0x22000000 = 0x22000000 + (0*32) + (0*4)$.
- The alias word at 0x2200001C maps to bit[7] of the bit-band byte at 0x20000000: $0x2200001C = 0x22000000 + (0*32) + (7*4)$.

Figure 9. Bit-band mapping



Directly accessing an alias region

Writing to a word in the alias region updates a single bit in the bit-band region.

Bit[0] of the value written to a word in the alias region determines the value written to the targeted bit in the bit-band region. Writing a value with bit[0] set to 1 writes a 1 to the bit-band bit, and writing a value with bit[0] set to 0 writes a 0 to the bit-band bit.

Bits[31:1] of the alias word have no effect on the bit-band bit. Writing 0x01 has the same effect as writing 0xFF. Writing 0x00 has the same effect as writing 0x0E.

Reading a word in the alias region:

- 0x00000000 indicates that the targeted bit in the bit-band region is set to zero
- 0x00000001 indicates that the targeted bit in the bit-band region is set to 1

Directly accessing a bit-band region

[Behavior of memory accesses on page 30](#) describes the behavior of direct byte, halfword, or word accesses to the bit-band regions.

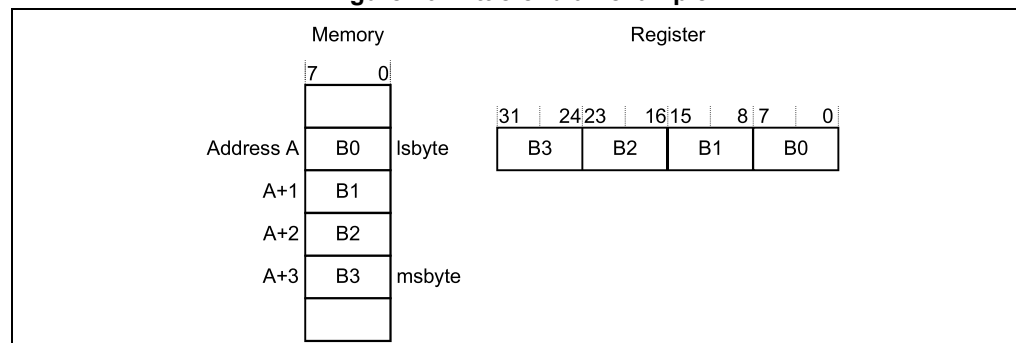
2.2.6 Memory endianness

The processor views memory as a linear collection of bytes numbered in ascending order from zero. For example, bytes 0-3 hold the first stored word, and bytes 4-7 hold the second stored word.

Little-endian format

In little-endian format, the processor stores the least significant byte of a word at the lowest-numbered byte, and the most significant byte at the highest-numbered byte. See [Figure 10](#) for an example.

Figure 10. Little-endian example



2.2.7 Synchronization primitives

The Cortex-M4 instruction set includes pairs of *synchronization primitives*. These provide a non-blocking mechanism that a thread or process can use to obtain exclusive access to a memory location. Software can use them to perform a guaranteed read-modify-write memory update sequence, or for a semaphore mechanism.

A pair of synchronization primitives comprises:

- Load-Exclusive instruction: used to read the value of a memory location, requesting exclusive access to that location.
- Store-Exclusive instruction: used to attempt to write to the same memory location, returning a status bit to a register. If this bit is:
 - 0: the thread or process gained exclusive access to memory, and the write succeeds.
 - 1: the thread or process did not gain exclusive access to memory, and no write is performed.